

Cx Dashboard

Automated HV/MV Substation Commissioning Tracker

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Audience	DCCD HV Team

1. Purpose

This memo presents the **Cx Dashboard** — a browser-based commissioning tracker that automates COR management, progress tracking, and analytics for HV/MV substation projects across DCCD. It covers the problem, the solution, quantified outcomes, and the path to broader adoption.

Developed during an internship on the ACx team in Dublin, under the mentorship of Ovais Tariq and programme direction of Sarath Ram S R.

2. Background

2.1 The ACx Team & DCCD HV Portfolio

The Amazon Commissioning (ACx) team oversees HV/MV substation commissioning across DCCD's global data centre portfolio. Current Procore data shows:

- **107** substation projects (51 EMEA, 35 APAC, 21 other regions)
- **1M+** total inspection records across all projects
- **39** unique Cx-prefixed inspection templates
- **101,581** CxHV inspections (65,796 open, 35,416 closed)
- Only **11 of 107** projects have item-level inspection data recorded

2.2 Commissioning Workflow

- 1. Scope Definition** — Define substation bays, equipment, and applicable tests.
- 2. SAT (Site Acceptance Testing)** — Contractor executes tests, records results.
- 3. CxA Witness / Review** — Commissioning Agent witnesses critical tests, signs off.
- 4. Close-Out** — All tests passed and witnessed; COR finalised.
- 5. COR Submission** — Completed COR submitted for programme record.

2.3 Current Tools

Procore manages inspection records and document control. Granular test tracking is done in Excel-based CORs and test scripts maintained per project. There is no consolidated tool bridging Procore's milestone-level view with the test-level detail required for commissioning sign-off.

3. Problem / Opportunity

3.1 Commissioning Data Landscape

Procore provides essential milestone tracking, but the commissioning data landscape has structural gaps that prevent test-level visibility:

- Procore line items are **intentionally high-level** — designed for milestone tracking (~19 items/equipment)
- Granular test data lives in **attached Excel test scripts**, not in Procore's structured fields
- Test scripts **don't cover all HV-specific tests** — some equipment types lack approved scripts
- DCCD team is **actively working with CxP** to get new test scripts approved
- No consolidated view aggregating test status **across a bay or substation**

Dimension	Procore Inspections	COR / Test Scripts
Granularity	~19 checklist items/equip	20–35 individual tests/equip
Data model	Pass/Fail milestone	SAT → CxA → Close-out per test
Coverage	39 templates across all projects	Project-specific, ad-hoc
Aggregation	Per-inspection only	No cross-bay/substation view
Items tracked	101,581 (inspection-level)	Unstructured (Excel files)

3.2 Manual COR Generation

CORs are built manually in Excel — copying test results across tabs, formatting sheets per bay, and reconciling data from multiple sources. A single 220kV substation COR takes 4–6 hours per revision.

- Copy-paste errors across 20+ column sheets
- No version control or audit trail
- Each revision restarts the manual process
- Multiple sub-CORs per project (e.g., ZAZ062 had 4 separate files)

3.3 No Real-Time Visibility

Progress data exists only in static spreadsheets shared intermittently. There is no way for programme managers to see live commissioning status across bays or projects.

3.4 Cross-Site Inconsistency

Each CxA maintains their own COR format and test lists. Without standardised templates, comparing progress across sites or regions is impractical.

4. Solution — Cx Dashboard

The Cx Dashboard is a browser-based application (React + Vite, zero backend) that provides end-to-end commissioning management — from scope definition to COR export. All data persists locally; no servers, databases, or API keys required.

Feature	What It Does	Why It Matters
Bay Builder	Visual scope builder — add bays, select equipment from a 56+ type palette, auto-populate tests.	Cuts scope definition from 1–2 hrs to 10–15 min. Ensures consistency across projects.
Progress Tracking	Per-test status tracking through SAT → CxA → Close-out stages with section/bay rollup.	Real-time visibility at any granularity — test, equipment, bay, or substation level.
Schedule	Date inputs per equipment with auto-scheduling. Status badges (Complete, In Progress, Delayed). Gantt view.	Replaces ad-hoc date tracking. Instant delay detection and programme oversight.
Analytics	Auto-generated charts: completion donut, section progress bars, Gantt overlay, bottleneck detection, velocity, health score.	Eliminates manual progress report compilation. Enables cross-project comparison.
COR Export	One-click Excel workbook generation matching DCCD COR template. Per-bay sheets, programme timeline, overview.	Reduces COR generation from 4–6 hrs to <5 min. Eliminates copy-paste errors.
COR Import	Load existing COR Excel files to auto-populate the dashboard. Handles format inconsistencies.	Enables migration of in-flight projects. No manual re-entry required.
Custom Templates	234+ standardised test templates. Create, modify, import/export equipment definitions.	Cross-team standardisation. Ensures no tests are missed regardless of CxA or region.
Asana Integration	Auto-generates Asana projects with sections (bays), task groups (equipment), and tasks (tests).	Enables task assignment, deadline tracking, and team collaboration through existing Asana workflows.

5. Results / Impact

5.1 Quantified Outcomes

Metric	Before (Manual)	After (Cx Dashboard)
COR generation (220kV sub)	4–6 hours / revision	< 5 minutes
Test-level visibility	None (inspection-level)	Real-time, per test
Tests tracked / substation	~19 Procore milestones	2,676 (ZAZ062)
Analytics	Manual compilation	Auto-generated
Template consistency	Varies by CxA / region	Standardised, reusable
Scope definition	1–2 hours	10–15 min (Bay Builder)
Equipment types supported	N/A (template-limited)	56+ types in palette
Test templates available	Ad-hoc per region	234+ standardised tests
COR sheets per export	Manual per bay	Auto (ZAZ062: 29 sheets)
Import existing COR	Re-entry from scratch	< 30 seconds

5.2 Proof of Concept — ZAZ062

Validated end-to-end on ZAZ062, a 220kV GIS substation with **26 bays**, **13 equipment types**, and **2,676 individual tests**. The full round-trip was confirmed: scope definition → progress tracking → COR export → COR import. The dashboard unified 4 separate sub-CORs (maintained by the assigned CxA) into a single consolidated view.

5.3 Portfolio Context & Projected Savings

Across the 107-project portfolio, only 11 have item-level data recorded — representing **100,000+ untracked tests**. If COR generation is reduced from ~5 hours to 5 minutes per revision across all projects, that yields **~530 hours saved per revision cycle**.

Each project typically requires ~3 revisions (initial scope, mid-programme update, final close-out). Total projected saving: **~1,600 hours** — equivalent to ~200 working days or nearly **1 FTE-year** redirected from manual spreadsheet work to commissioning activities. This excludes additional gains from automated analytics, schedule tracking, and reduced copy-paste errors.

Appendix: Dashboard Screenshots

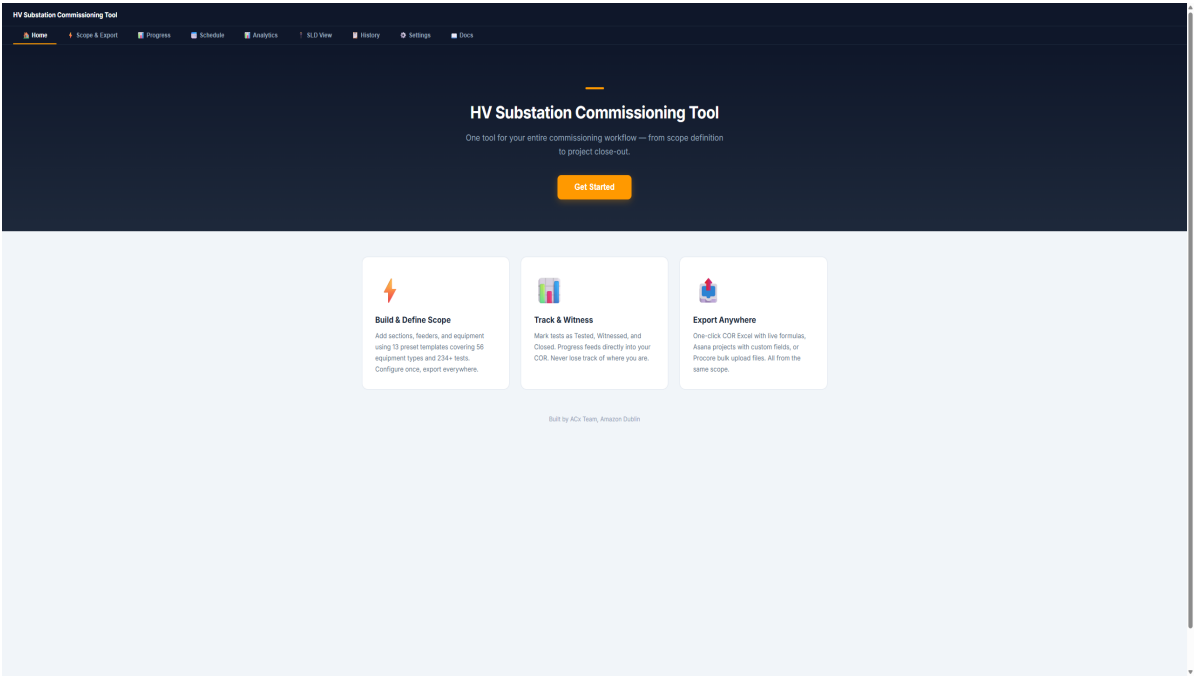


Figure 1 — Landing Page

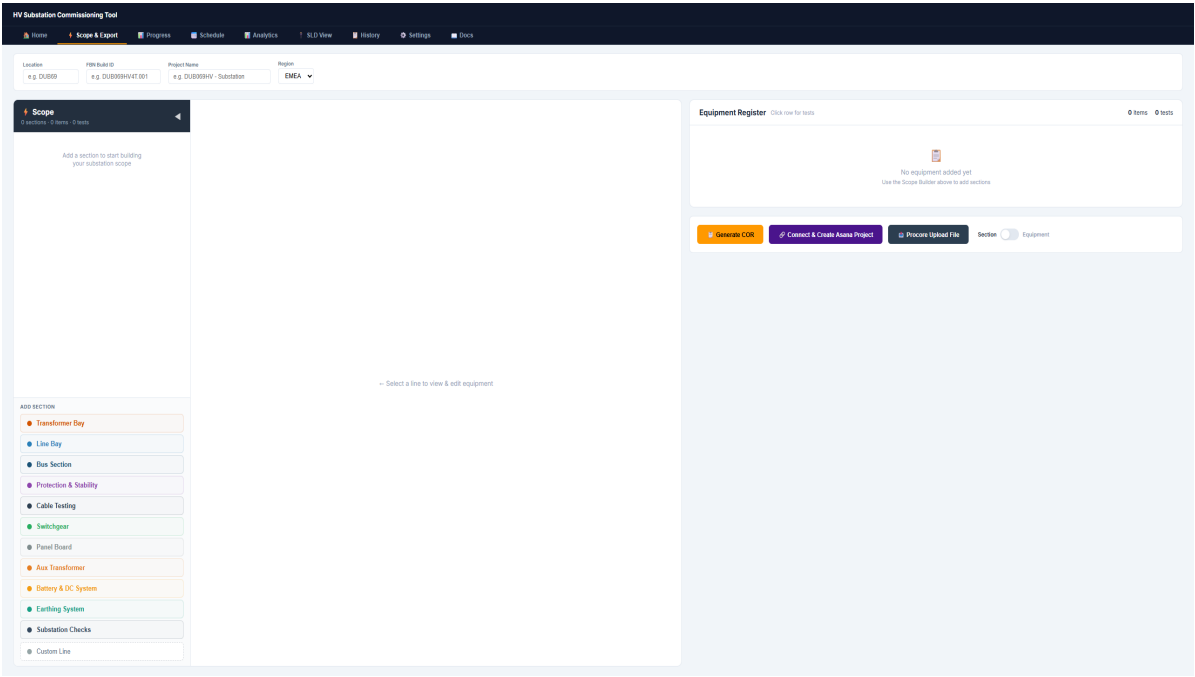


Figure 2 — Bay Builder (Scope & Export)

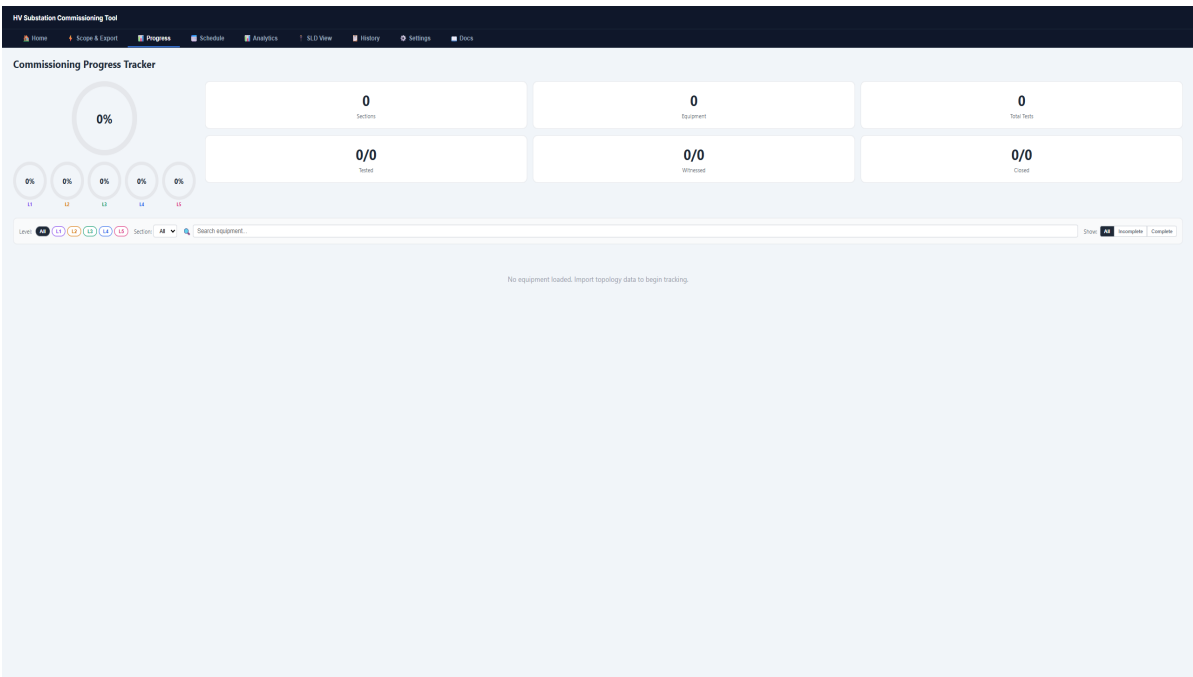


Figure 3 — Progress Tracker

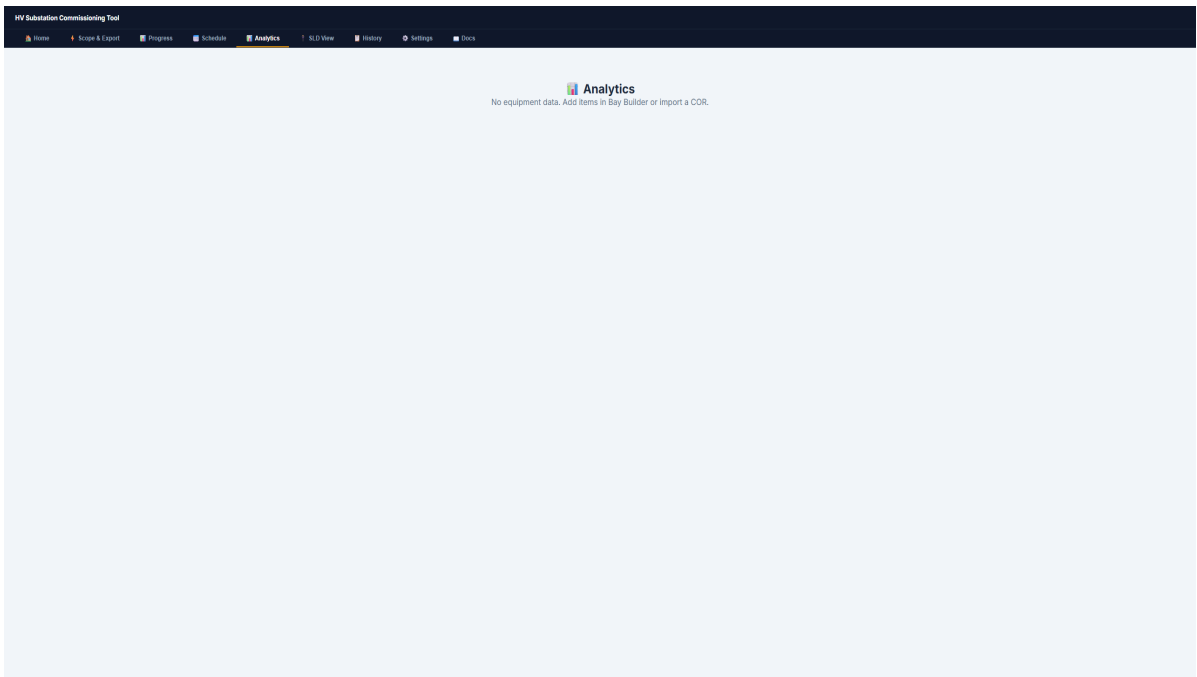


Figure 4 — Analytics & Metrics Dashboard

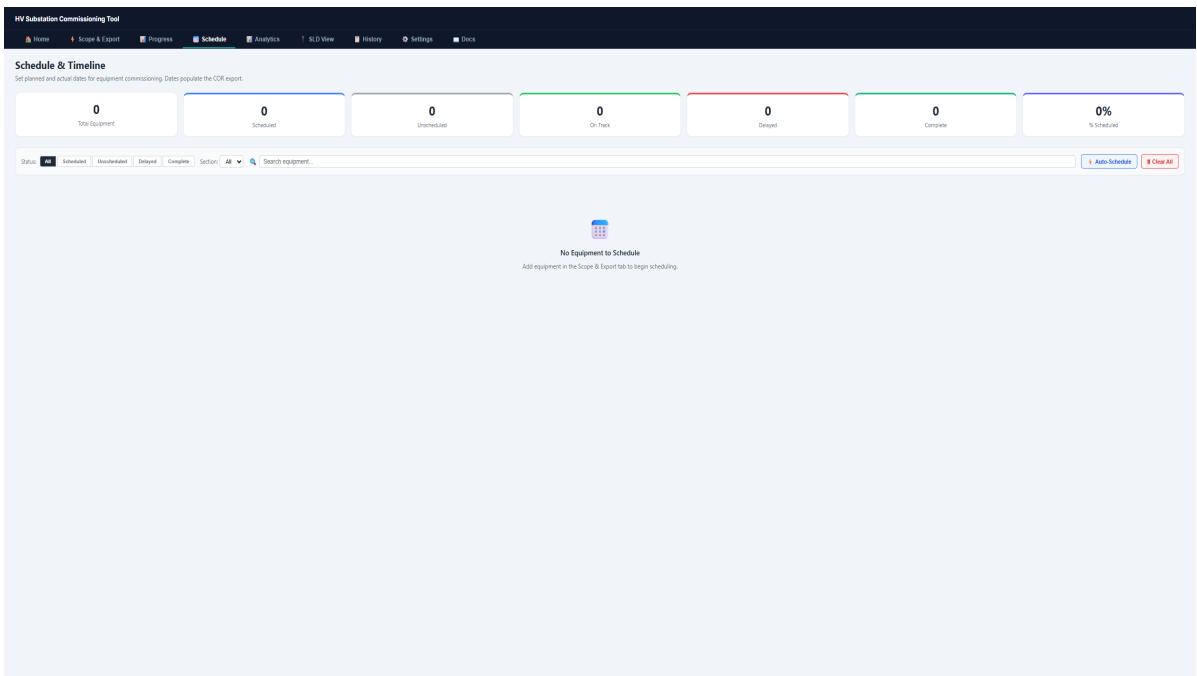


Figure 5 — Schedule & Timeline

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